Paper Presentation

SOLA-GCL: Subgraph-Oriented Learnable Augmentation for Graph Contrastive Learning

Tianhao Peng, Xuhong Li, Haitao Yuan, Yuchen Li, Haoyi Xiong

Presented By : Divyansh Jha (2022B3A70438P)

Made with GAMMA

Background & Motivation

Graph Contrastive Learning

Graph contrastive learning (GCL) has emerged as a powerful technique for learning robust and discriminative graph representations without labeled data. Traditional approaches generate augmented views through random perturbations, but often overlook the critical role of subgraph structures.

The Challenge

Existing methods struggle to capture **intra-subgraph characteristics** and **inter-subgraph relationships**—crucial features that vary significantly across different graph types like social networks and molecular structures.

Made with GAMMA

Limitations of Existing Methods

Loss of Subgraph Information

Graphs contain important details within subgraphs and between different subgraphs. Previous studies haven't effectively captured both intra-subgraph characteristics and inter-subgraph relationships.

Poor Adaptability

Different graph structures require distinct augmentation strategies. Uniform strategies across diverse datasets limit flexibility and effectiveness.

Semantic Loss

Random view generation processes may alter the original graph's semantics, changing critical structural information during augmentation.

Made with GAMMA

SOLA-GCL Framework Architecture

Graph Partitioning

Divide graph into densely connected subgraphs using algorithms like Louvain or RDKit

Augmentation Selector

Learn optimal augmentation strategy distribution for each subgraph using GNN embeddings

View Generator

Generate augmented subgraph views using selected strategies

View Assembler

Construct final augmented graph view by combining subgraph views

Made with GAMMA

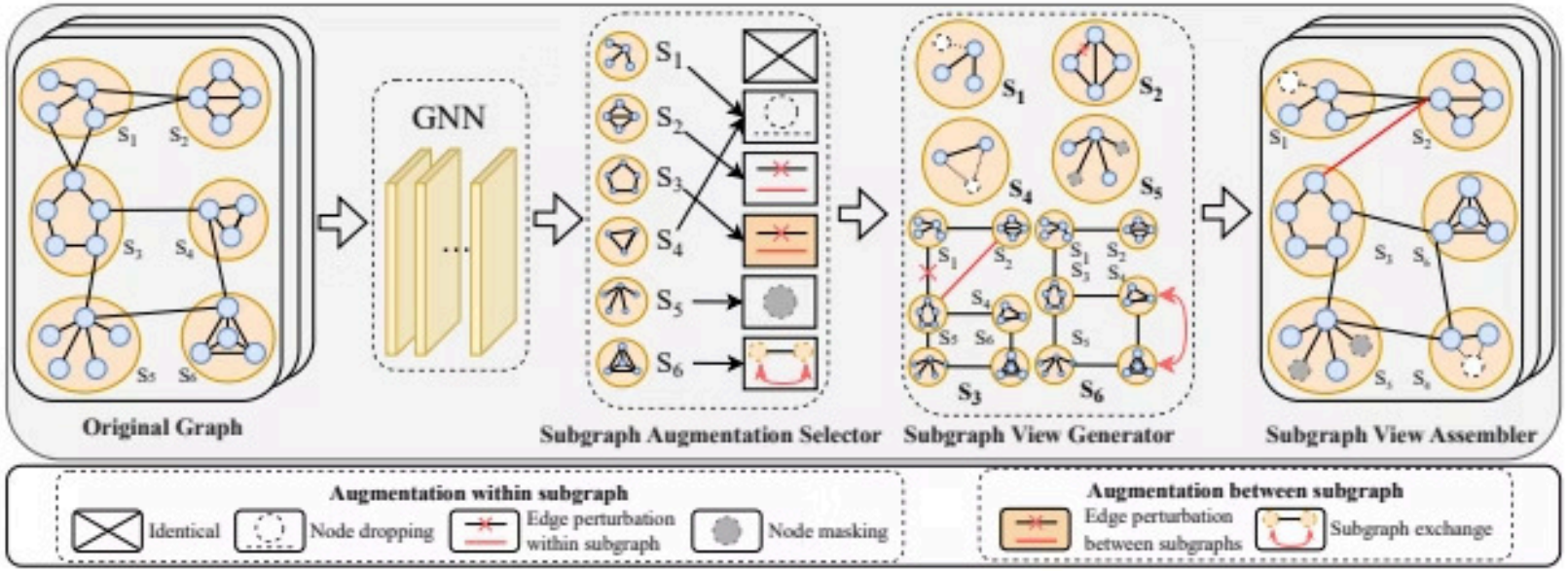

Figure 1: An illustration of the proposed SOLA-GCL framework. The graph view generator is composed of three critical components: the subgraph augmentation selector, the subgraph view generator, and the subgraph view assembler. The subgraph augmentation selector learns to choose the optimal augmentation strategy for each subgraph, and the subgraph view generator outputs augmented subgraph views according to the selected strategies. The subgraph view assembler constructs an augmented graph view based on these augmented subgraph views.

Made with GAMMA

Augmentation Strategies

Intra-Subgraph Strategies

- **Node Dropping:** Conditioned on node representations to decide which nodes to drop
- **Feature Masking:** Mask node features within subgraphs
- **Intra-Edge Perturbation:** Add or remove edges within subgraphs

Inter-Subgraph Strategies

- **Inter-Edge Perturbation:** Perturb edges between different subgraphs
- **Subgraph Swapping:** Swap positions by changing inter-subgraph edges

Each strategy is applied in a learnable manner using Gumbel-Softmax for differentiable sampling, preserving semantic integrity while enabling adaptive augmentation.

Made with GAMMA

Contrastive Learning in SOLA-GCL

01

Generate Augmented Views

Create two augmented views by perturbing the graph using learnable subgraph-oriented strategies

02

Compute Representations

Pass augmented views through GNN to obtain graph embeddings

03

Maximize Consistency

Maximize feature consistency between positive pairs using contrastive loss

04

Minimize Similarity

Minimize mutual information between augmentation selections and views

Made with GAMMA

Experimental Results: Unsupervised Learning

Model	PROTEINS	DD	NCI1	COLLAB
Full Data	78.25±1.61	80.73±3.78	83.65±1.16	83.44±0.77
10% Data	69.72±6.71	74.36±5.86	75.16±2.07	74.34±2.00
10% GAE	70.51±0.17	74.54±0.68	74.36±0.24	75.09±0.19
10% Infomax	72.27±0.40	75.78±0.34	74.86±0.26	73.76±0.29
10% ContextPred	70.23±0.63	74.66±0.51	73.00±0.30	73.69±0.37
10% GCA	73.85±5.56	76.74±4.09	68.73±2.36	74.32±2.30
10% GraphCL	74.21±4.50	76.65±5.12	73.16±2.90	75.50±2.15
10% JOAO	72.13±0.92	75.69±0.67	74.48±0.27	75.30±0.32
10% JOAOv2	73.31±0.48	75.81±0.73	74.86±0.39	75.53±0.18
10% AD-GCL	73.96±0.47	77.91±0.73	**75.18±0.31**	75.82±0.26
10% AutoGCL	75.65±2.40	77.50±4.41	73.75±2.25	**77.16±1.48**
10% SimGRACE	74.03±0.51	76.48±0.52	74.60±0.41	74.74±0.28
10% MSSGCL	75.76±0.52	78.89±0.18	74.77±0.31	76.02±0.13
10% **Ours**	**77.63±3.10**	**79.43±4.31**	74.94±1.96	76.97±1.31

Table 3: Comparison with existing methods and different strategies for semi-supervised learning.

	MUTAG	NCI1	COLLAB	IMDB-B
Louvain	90.49±2.22	82.07±1.32	73.80±1.32	74.20±1.89
GN	89.88±2.98	81.56±2.11	72.32±1.88	74.10±2.84
SOLA-GCL-P	88.33±2.04	79.93±1.74	70.92±1.91	71.70±2.80
SOLA-GCL-S	88.80±3.46	81.58±1.57	71.86±1.63	74.00±2.63
SOLA-GCL-R	88.80±3.24	80.95±1.75	69.38±1.96	73.70±2.84

Table 4: Ablation study on the critical modules.

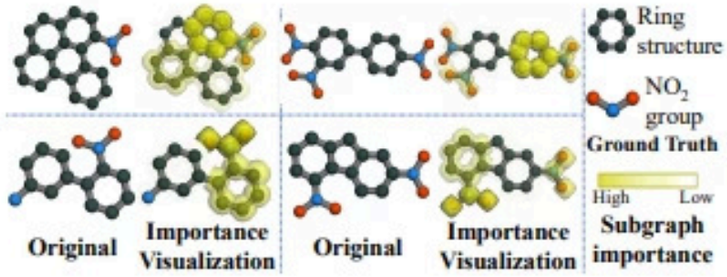

Figure 2: We visualize the graphs in the MUTAG dataset, the deeper color indicates the more important subgraphs and the ground truth is ring structure and NO_2 group.

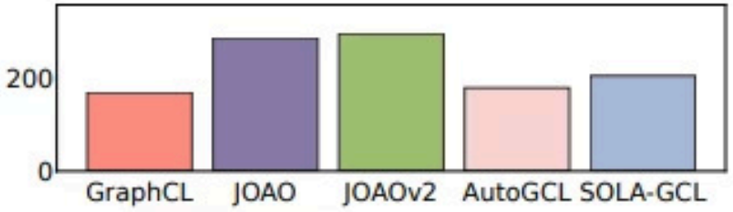

Figure 3: Comparisons in terms of real running time. Each model is trained for 100 epochs on each dataset and the average training time of one epoch is reported.

90.49%

MUTAG

Best performance on molecular dataset

82.07%

NCI1

Chemical compound classification

73.80%

COLLAB

Scientific collaboration network

74.20%

IMDB-B

Movie collaboration network

SOLA-GCL consistently outperformed state-of-the-art methods across 8 datasets, demonstrating superior performance in unsupervised graph classification tasks.

Made with GAMMA

Results Across Learning Settings

Transfer Learning

Achieved SOTA or second-best performance on 6 molecular datasets (BBBP, ToxCast, SIDER, ClinTox, HIV, BACE), demonstrating robustness in pre-training scenarios

Semi-Supervised Learning

Consistently achieved best performance on PROTEINS, DD, NCI1, and COLLAB with limited labeled data, showing effectiveness with scarce annotations

Efficiency Analysis

Strikes optimal balance between efficiency and effectiveness, consuming less training time than JOAO/JOAOv2 while achieving better performance

Made with GAMMA

Key Contributions & Conclusion

Novel Framework

First work to build learnable generative augmentation policies focusing specifically on intra-subgraph characteristics and inter-subgraph relationships

End-to-End Training

Introduced differentiable training algorithm enabling automatic augmentation strategy selection and graph view generation

Extensive Validation

Demonstrated robustness across semi-supervised, unsupervised, and transfer learning settings on diverse graph classification datasets

SOLA-GCL effectively captures subgraph information through learnable augmentation strategies, consistently outperforming existing methods while preserving semantic integrity across various graph types and learning scenarios.

Made with GAMMA